

LEIPZIG HALLE, Germany

WMO: 104690

Lat: 51.4347N

Lon: 12.2397E

Elev: 445

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	17.5	7.0	8.0	14.2	12.3	10.5	18.2	32.3	44.5	29.1	43.5	8.8	70	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	88.1	67.3	84.4	66.2	80.8	65.2	69.8	81.9	68.3	79.9	66.9	77.6	8.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	98.6	72.7	64.6	92.7	71.5	62.9	87.1	70.8	34.1	82.4	32.9	79.6	31.8	77.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	22.7	20.1	DB	8.5	94.7	7.7	3.2	2.9	97.1	-1.6	98.9	-5.9	100.8	-11.5	103.1	(4)
(5)			WB	7.8	73.3	7.6	1.8	2.4	74.6	-2.1	75.6	-6.4	76.7	-11.9	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	33.2	34.2	41.1	50.6	57.2	64.0	68.3	66.5	60.0	51.2	42.7	37.1	(6)
(7)		DBStd	14.13	8.93	8.32	8.10	7.18	7.10	6.49	6.14	5.75	5.85	6.74	6.87	7.76	(7)
(8)		HDD50	2054	522	442	287	80	16	1	0	0	1	67	236	402	(8)
(9)		HDD65	5597	986	862	741	433	260	94	37	48	171	429	670	866	(9)
(10)		CDD50	2268	1	0	11	98	238	420	566	511	302	105	15	1	(10)
(11)		CDD65	338	0	0	0	1	17	64	138	94	23	1	0	0	(11)
(12)		CDH74	3059	0	0	0	17	179	596	1268	797	196	6	0	0	(12)
(13)		CDH80	973	0	0	0	3	39	170	458	265	38	0	0	0	(13)
(14)	Wind	WSAvg	10.1	12.2	10.9	11.0	9.6	9.4	8.7	9.0	8.7	9.1	9.7	10.7	12.1	(14)
(15)	Precipitation	PrecAvg	21.9	1.4	1.1	1.5	1.3	2.2	2.3	3.0	2.4	2.0	1.4	1.8	1.5	(15)
(16)		PrecMax	30.7	2.4	2.1	3.7	3.3	5.1	3.6	5.3	5.9	5.4	3.7	4.1	3.1	(16)
(17)		PrecMin	14.3	0.2	0.2	0.2	0.1	0.2	0.6	0.6	0.9	0.5	0.3	0.0	0.6	(17)
(18)		PrecStd	3.6	0.6	0.5	0.9	0.7	1.3	0.8	1.2	1.2	1.1	0.8	0.9	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	55.7	66.6	75.6	84.5	89.6	94.8	93.1	84.5	74.8	62.6	55.4	(19)
(20)			MCWB	49.8	46.1	53.4	59.1	66.0	68.5	69.4	67.8	64.9	62.0	57.0	50.2	(20)
(21)		2%	DB	51.4	51.5	60.7	71.6	78.9	84.5	89.4	86.4	79.2	68.1	57.6	51.9	(21)
(22)			MCWB	47.5	46.2	51.3	57.2	63.4	66.5	67.2	68.2	63.1	59.6	53.3	48.1	(22)
(23)		5%	DB	48.1	48.3	55.7	68.0	74.8	80.8	85.6	82.2	75.2	64.5	55.3	49.9	(23)
(24)			MCWB	45.3	44.0	48.2	56.1	61.6	65.6	66.5	65.9	62.7	57.7	51.4	46.5	(24)
(25)		10%	DB	44.9	45.0	52.2	64.2	70.2	76.9	80.8	77.4	71.2	61.1	52.1	46.8	(25)
(26)			MCWB	42.7	41.4	46.4	54.3	59.0	63.5	64.9	64.1	61.1	56.3	48.7	44.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.0	49.3	54.9	60.8	68.2	72.6	71.4	72.2	68.0	63.2	57.5	51.3	(27)
(28)			MCDWB	54.2	51.7	62.8	72.2	80.6	82.6	85.9	85.3	79.5	71.0	61.4	54.6	(28)
(29)		2%	WB	47.8	46.5	52.0	58.7	65.3	68.7	69.7	69.5	65.4	60.4	54.3	48.9	(29)
(30)			MCDWB	50.7	50.5	58.5	69.0	76.2	80.4	82.3	81.2	74.7	66.2	57.7	51.6	(30)
(31)		5%	WB	45.0	44.0	49.7	57.0	62.8	66.7	68.2	67.8	63.5	58.5	51.8	46.6	(31)
(32)			MCDWB	47.2	47.7	54.8	66.6	72.3	77.7	79.8	78.8	72.1	63.5	55.0	49.3	(32)
(33)		10%	WB	42.6	41.5	47.2	54.9	60.3	64.7	66.9	66.1	61.7	56.6	49.6	44.5	(33)
(34)			MCDWB	44.5	44.7	52.0	62.5	68.2	74.4	77.6	74.9	69.1	61.1	52.1	46.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	10.9	14.1	18.1	17.9	18.3	18.4	17.8	16.4	13.6	9.8	8.6	(35)
(36)			MCDBR	10.6	15.2	20.9	24.5	24.2	24.5	25.2	25.1	23.2	17.8	12.9	10.8	(36)
(37)			MCWBR	8.4	10.5	12.8	13.2	11.6	10.5	8.8	9.5	10.4	10.5	9.4	8.6	(37)
(38)		5% WB	MCDWB	10.0	14.1	19.0	22.5	22.2	22.2	20.7	21.0	19.4	16.3	11.9	10.1	(38)
(39)			MCWBR	8.3	10.2	12.3	12.7	11.1	10.4	8.3	9.4	10.0	10.4	9.2	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.341	0.384	0.417	0.419	0.415	0.433	0.426	0.388	0.366	0.340	0.311	(40)	
(41)		taud	2.362	2.329	2.223	2.184	2.215	2.255	2.212	2.251	2.338	2.373	2.385	2.364	(41)	
(42)		Ebn at Noon	220	246	254	259	265	267	259	254	251	233	208	201	(42)	
(43)		Edh at Noon	20	27	36	42	43	41	42	39	32	26	20	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	233	452	776	1266	1548	1658	1565	1352	974	566	268	178	(44)	
(45)		RadStd	30	73	98	160	184	137	199	121	123	65	37	20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A

Nomenclature: See separate page